# Data Analysis & Insights Report

## 1. Dataset Overview

- **Total Rows:** 1,060,137
- **Total Columns:** 11
- **Numeric Features (9):** GROUPE, CODE, OUVERTURE, CLOTURE, PLUS_BAS, PLUS_HAUT, QUANTITE_NEGOCIEE, NB_TRANSACTION, CAPITAUX
- **Categorical Features (2):** SEANCE, VALEUR

## 2. Data Quality & Missing Values

- **SEANCE:** 4 missing entries (0.00%)
- **GROUPE:** 85,132 missing entries (8.03%)
- **CODE:** 914,852 missing entries (86.30%)
- **VALEUR:** 85,130 missing entries (8.03%)
- **OUVERTURE:** 83,929 missing entries (7.92%)
- **CLOTURE:** 4 missing entries (0.00%)
- **PLUS_BAS:** 4 missing entries (0.00%)
- **PLUS_HAUT:** 4 missing entries (0.00%)
- **QUANTITE_NEGOCIEE:** 4 missing entries (0.00%)
- **NB_TRANSACTION:** 85,132 missing entries (8.03%)
- **CAPITAUX:** 7,192 missing entries (0.68%)

## 3. Descriptive Statistics (Numeric)

**GROUPE:** Mean = 25.65, Median = 21.0, Min = 11.0, Max = 99.0, StdDev = 13.59

**CODE:** Mean = 469,160.62, Median = 462,014.0, Min = 100,010.0, Max = 783,001.0, StdDev = 217,578.62

**OUVERTURE:** Mean = 394.04, Median = 7.35, Min = 0.0, Max = 300,617.0, StdDev = 9,209.87

**CLOTURE:** Mean = 35.74, Median = 5.54, Min = 0.0, Max = 242.42, StdDev = 44.91

**PLUS_BAS:** Mean = 2.59, Median = 0.0, Min = 0.0, Max = 242.42, StdDev = 12.17

**PLUS_HAUT:** Mean = 2.68, Median = 0.0, Min = 0.0, Max = 242.42, StdDev = 12.19

**QUANTITE_NEGOCIEE:** Mean = 2,048.75, Median = 0.0, Min = 0.0, Max = 15,018,924.0, StdDev = 52,447.69

**NB_TRANSACTION:** Mean = 238.93, Median = 0.0, Min = 0.0, Max = 45,862,421.0, StdDev = 47,775.3

**CAPITAUX:** Mean = 18,008.44, Median = 0.0, Min = 0.0, Max = 183,300,000.0, StdDev = 568,242.95

## 4. Categorical Features Summary

- **SEANCE** (2,508 unique values)

  *Top values:* 2023-01-05 [1,192], 2023-01-02 [1,190], 2023-01-03 [1,190]

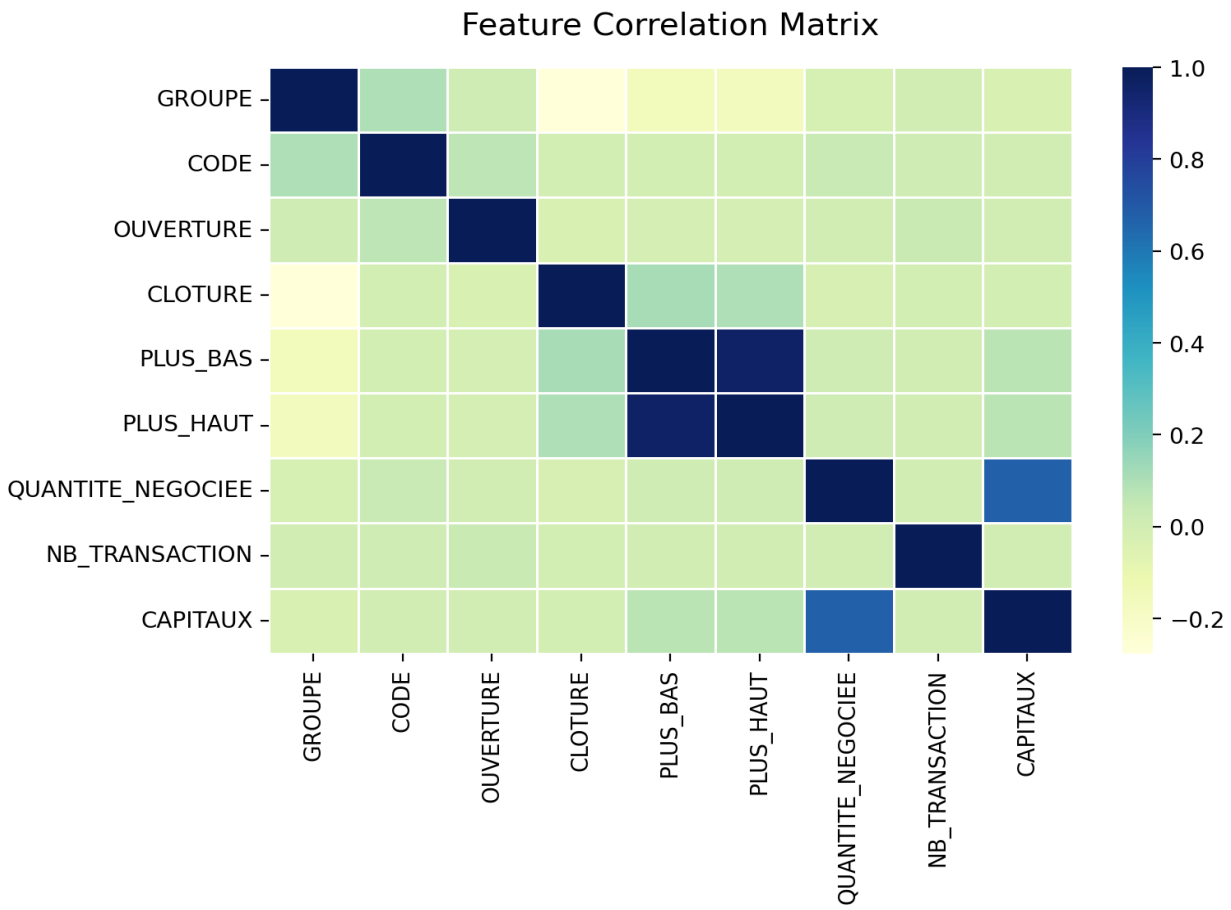
- **VALEUR** (928 unique values)

  *Top values:* SIAME [4,778], STB [3,494], MONOPRIX [3,272]

## 5. Data Visualizations

### Correlation Heatmap

This matrix displays the linear correlation coefficients between numeric features. Values closer to 1 or -1 indicate strong relationships.

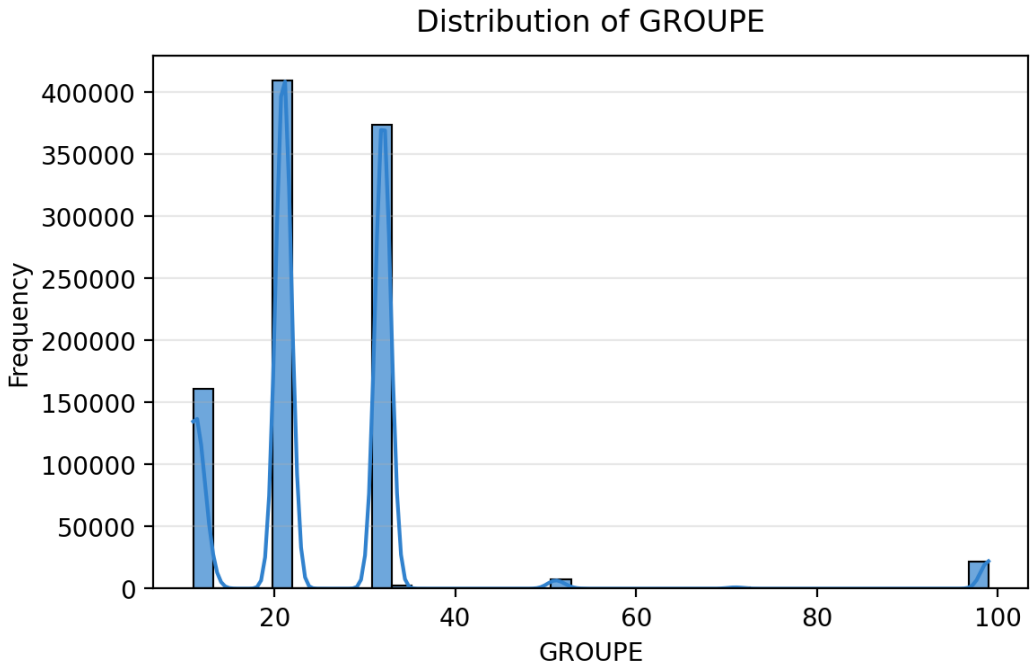


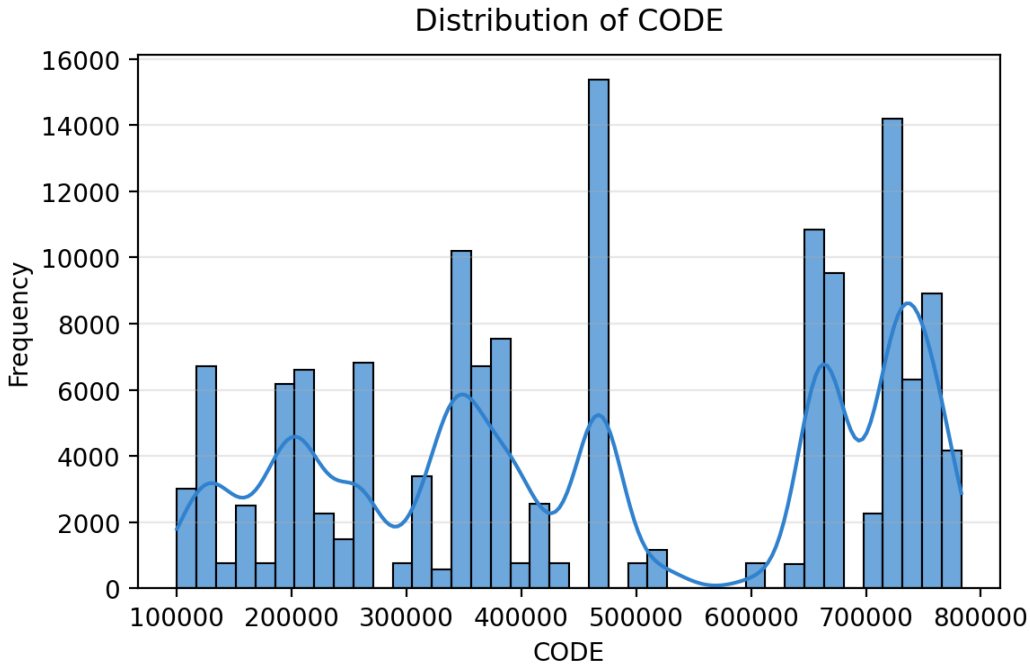

### Top Correlations:

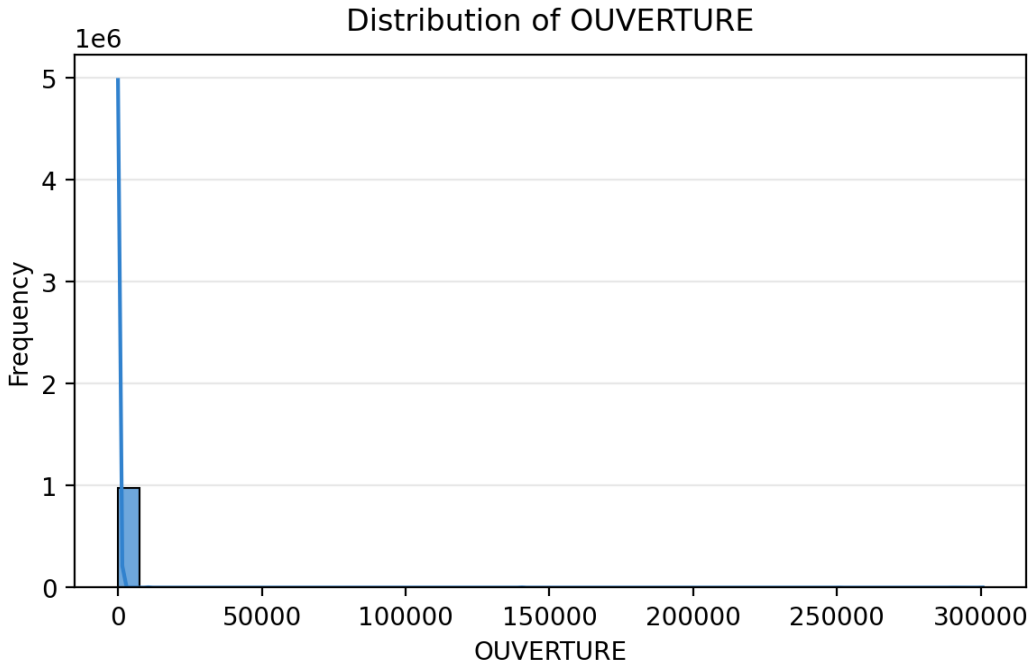
- **PLUS_BAS & PLUS_HAUT:** 0.962 (Positive)
- **QUANTITE_NEGOCIEE & CAPITAUX:** 0.676 (Positive)
- **GROUPE & CLOTURE:** -0.276 (Negative)
- **GROUPE & PLUS_HAUT:** -0.162 (Negative)
- **GROUPE & PLUS_BAS:** -0.160 (Negative)

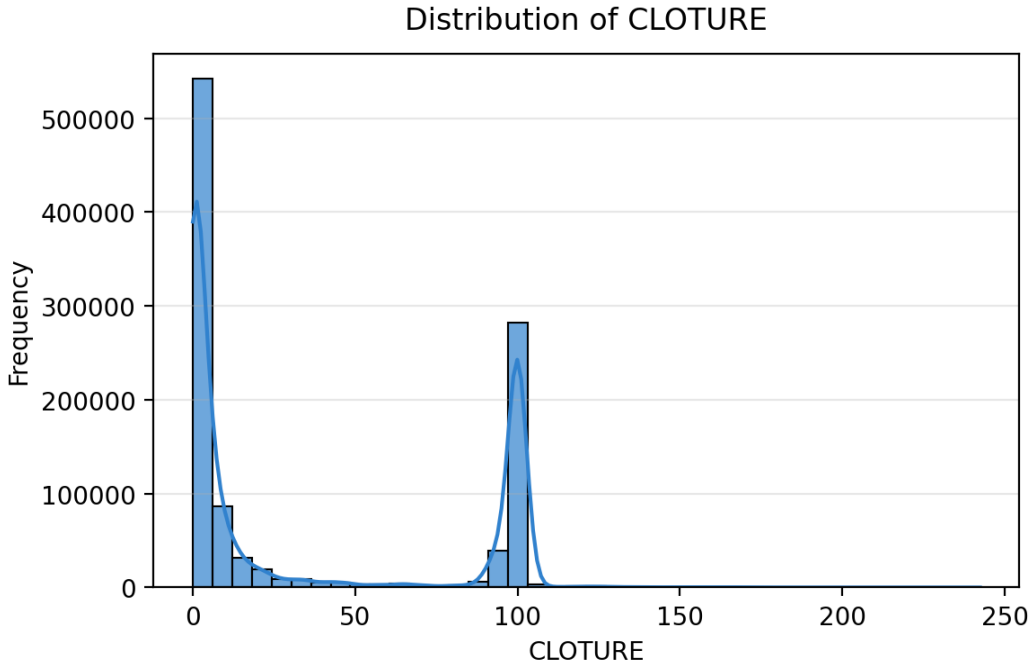
## Numeric Feature Distributions

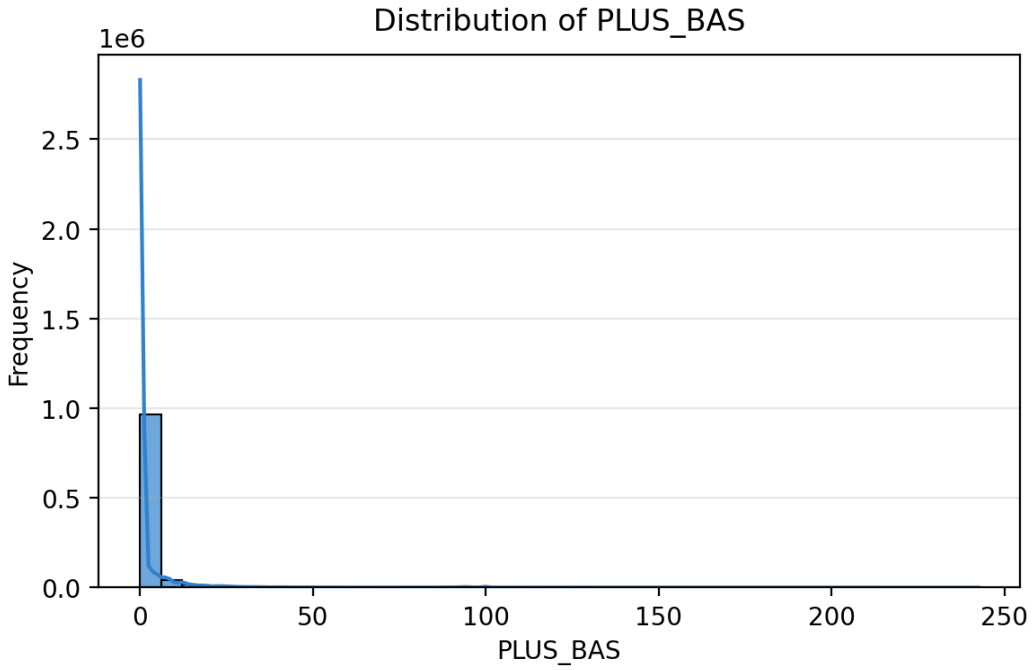
Histograms showing the underlying distribution and spread of numerical data.

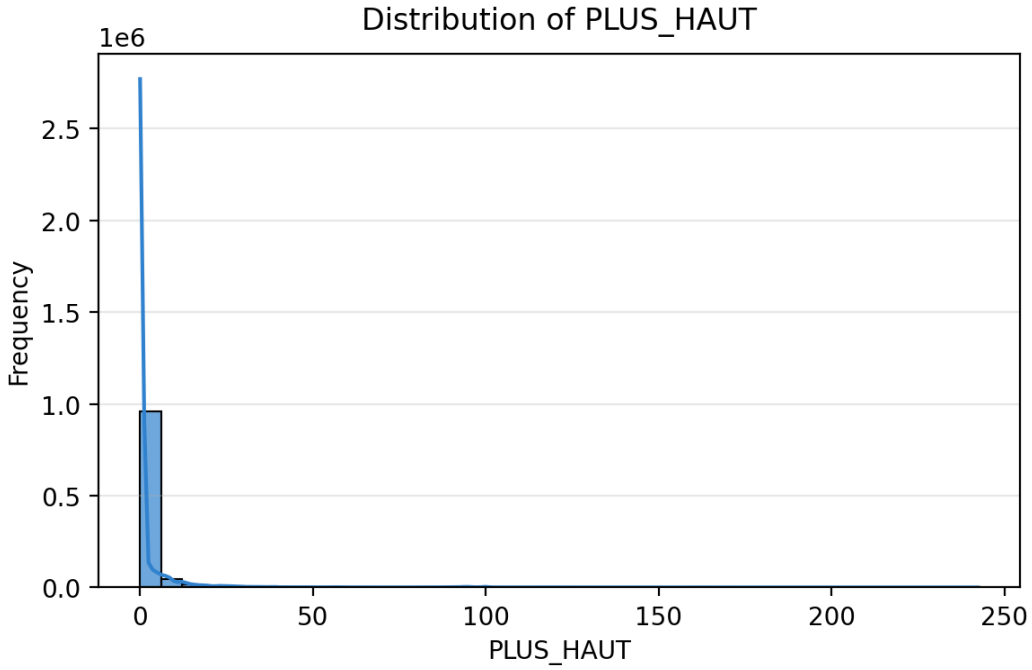


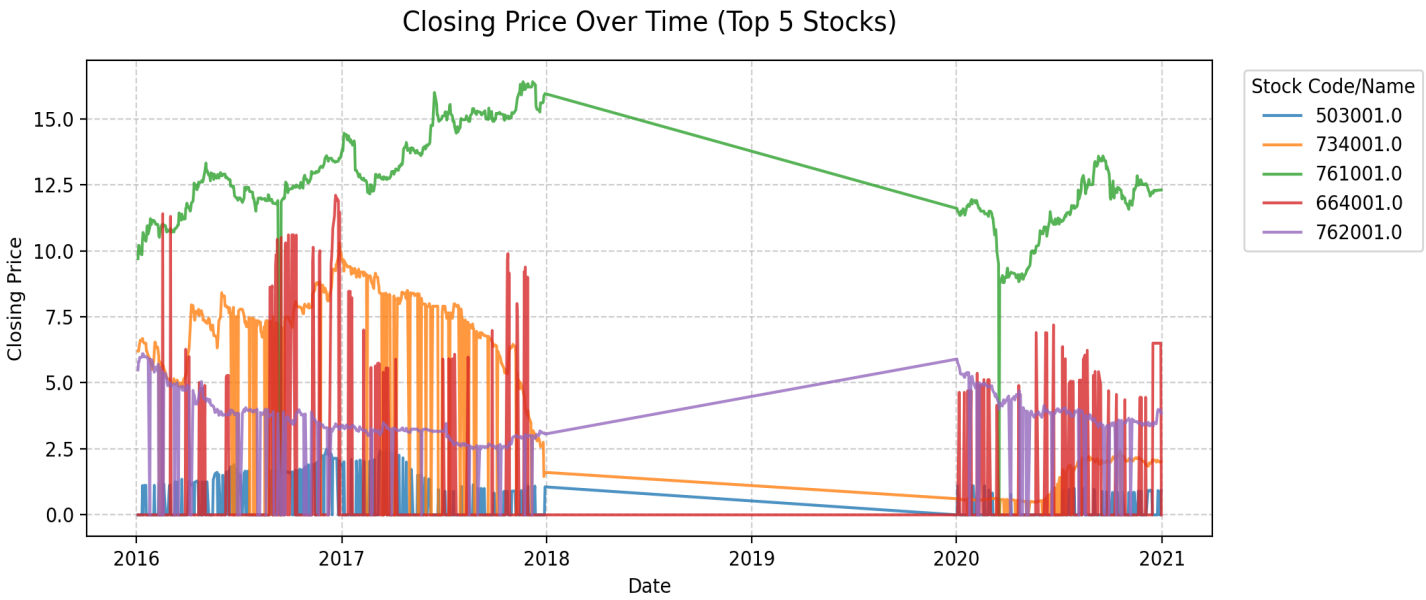




Distribution of OUVERTURE
1e6
5
4
3
2
1
0
Frequency
0
50000
100000
150000
200000
250000
300000
OUVERTURE


Distribution of CLOTURE
500000
400000
300000
200000
100000
0
Frequency
0
50
100
150
200
250
CLOTURE


Distribution of PLUS_BAS
1e6
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
0
50
100
150
200
250
PLUS_BAS


Distribution of PLUS_HAUT
1e6
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
0
50
100
150
200
250
PLUS_HAUT

## Time-Series Analysis (Top Traded Stocks)

Closing price history for the 5 most frequently recorded stocks.